LAB 17 – Decision Tree Depth Analysis

Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_
from sklearn.preprocessing import LabelEncoder
```

Load Dataset

```
df = pd.read_csv("/content/data.csv")

df.head()
```

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
0	842302	M	17.99	10.38	122.80	1001.0
1	842517	M	20.57	17.77	132.90	1326.0
2	84300903	M	19.69	21.25	130.00	1203.0
3	84348301	M	11.42	20.38	77.58	386.1
4	84358402	M	20.29	14.34	135.10	1297.0

5 rows × 33 columns

Drop Unnecessary Columns

```
df = df.drop(['id', 'Unnamed: 32'], axis=1)
```

Encode Target (M=1, B=0)

```
le = LabelEncoder()

df['diagnosis'] = le.fit_transform(df['diagnosis'])

df.head()
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness
0	1	17.99	10.38	122.80	1001.0	0.
1	1	20.57	17.77	132.90	1326.0	0.
2	1	19.69	21.25	130.00	1203.0	0.
3	1	11.42	20.38	77.58	386.1	0.
4	1	20.29	14.34	135.10	1297.0	0.

5 rows × 31 columns

Define Features and Target

```
X = df.drop('diagnosis', axis=1)
y = df['diagnosis']
```

Train/Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Train Decision Trees with Different Depths

```
depths = [3, 5, 7, 10]

results = []

for d in depths:
    model = DecisionTreeClassifier(max_depth=d, random_state=42)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    results.append([d, acc, prec, rec, f1])

results_df = pd.DataFrame(
    results,
    columns=["Max Depth", "Accuracy", "Precision", "Recall", "F1 Score"]
```

```
)

results_df
```

	Max Depth	Accuracy	Precision	Recall	F1 Score
0	3	0.947368	0.951220	0.906977	0.928571
1	5	0.947368	0.930233	0.930233	0.930233
2	7	0.947368	0.930233	0.930233	0.930233
3	10	0.947368	0.930233	0.930233	0.930233

Next steps: Generate code with `results_df` New interactive sheet

Display Results Clearly

```
print("Decision Tree Depth Analysis Results")
print(results_df)
```

```
Decision Tree Depth Analysis Results
   Max Depth  Accuracy  Precision    Recall  F1 Score
0          3  0.947368   0.951220  0.906977  0.928571
1          5  0.947368   0.930233  0.930233  0.930233
2          7  0.947368   0.930233  0.930233  0.930233
3         10  0.947368   0.930233  0.930233  0.930233
```

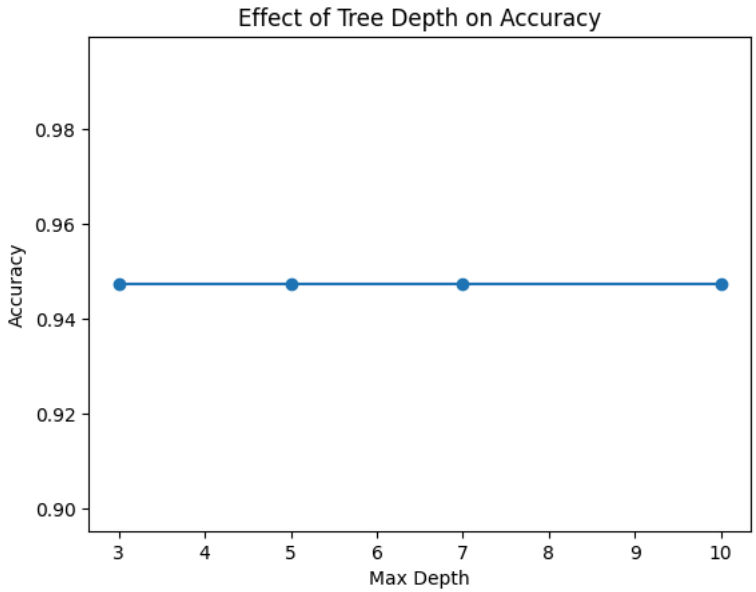
Plot Depth vs Accuracy

```
import matplotlib.pyplot as plt

plt.plot(results_df["Max Depth"], results_df["Accuracy"], marker='o')

plt.xlabel("Max Depth")
plt.ylabel("Accuracy")
plt.title("Effect of Tree Depth on Accuracy")

plt.show()
```


Effect of Tree Depth on Accuracy
Accuracy
0.98
0.96
0.94
0.92
0.90
3
4
5
6
7
8
9
10
Max Depth